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Politecnico di Milano — MSc in High Performance Computing Engineering

WORKLOAD-DRIVEN EXPLORATION OF EDGE NPU ARCHITECTURES FOR VISION-LANGUAGE-ACTION MODELS

PROGRESS MEETING #00 — PROJECT KICKOFF & SCOPE

Alessandro Ghiotto

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OUTLINE



1 Foundations of DNN accelerators

TEMPORAL VS. SPATIAL ARCHITECTURES

► Temporal (SIMD/SIMT):

- Centralized control logic.
- ALUs fetch/store only via memory hierarchy.
- No direct communication between ALUs.

► Spatial (Dataflow):

- Distributed PEs with local scratchpads.
- Direct inter-PE communication.
- Enables *spatial reuse* and dataflow optimizations.

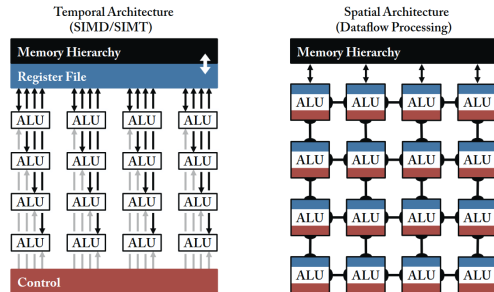


Figure: Temporal vs. Spatial Architecture

DATAFLOWS & NOC CONNECTIVITY

- **Dataflow Taxonomy:** Loop-nest ordering prioritizes stationarity (*WS*, *OS*, *IS*, or *RS*) to minimize memory access energy.
- **NoC Connectivity:** Trade-off between hardware bandwidth cost and spatial data reuse capability.

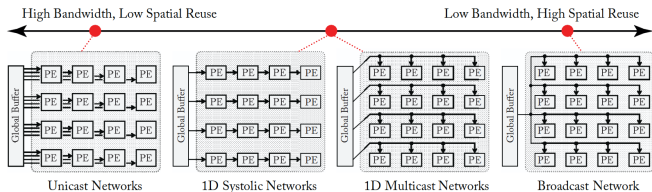


Figure: NoC Topologies: Bandwidth vs. Spatial Reuse Trade-off

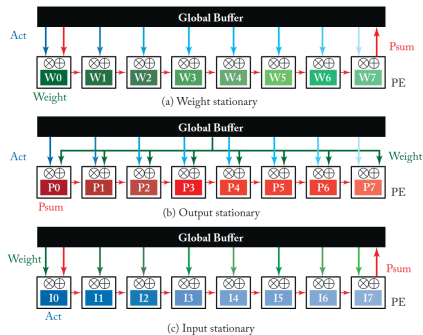


Figure: Dataflow architectures

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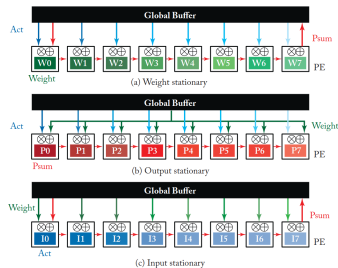


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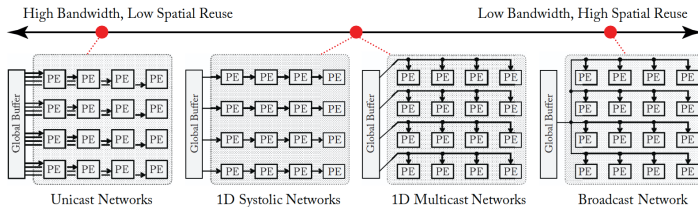


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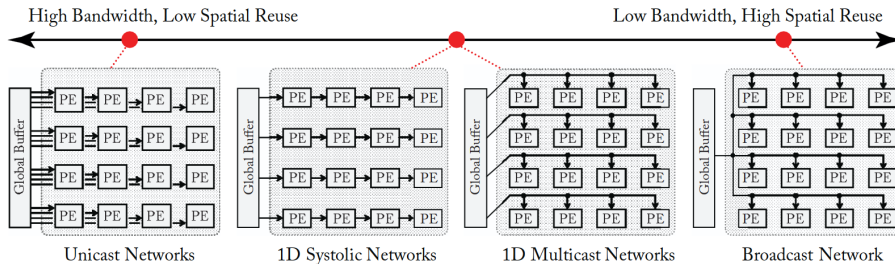


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